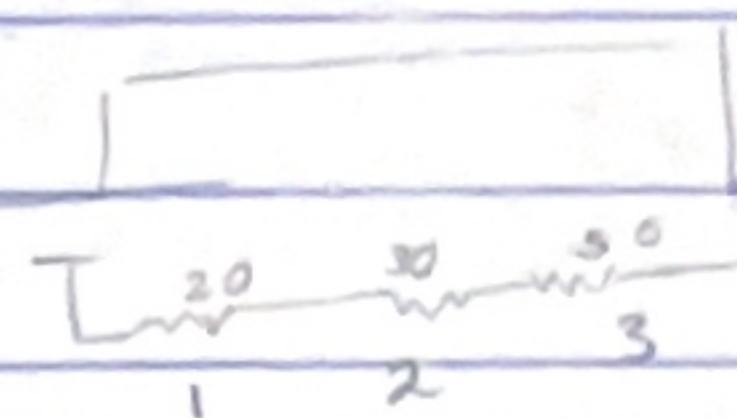


Embedded Systems & IOT

Part A

166082 ICS 4.1B



a)

1. a) Total Resistance in ohms

For circuit in series, $R_T = R_1 + R_2 + R_3$

$$= 20 + 30 + 50$$

$$R_T = 100\Omega$$

2) Total Current from Power supply

$$I = \frac{V}{R}, \text{ where } V = 125$$

$$R = 100$$

$$= \frac{125}{100}$$

$$I_T = 1.25A$$

3) Current through each resistor

$I_T = I_1 = I_2 = I_3 = I_4$ for a circuit in series

$I_T = 1.25A$, therefore current through each = $1.25A$

4) Voltage drop through each resistor

$$V = IR$$

$$\text{For } 20\Omega, 1.25 \times 20$$

$$= 25V$$

$$\text{For } 30\Omega, 1.25 \times 30$$

$$= 37.5V$$

$$\text{For } 50\Omega, 1.25 \times 50$$

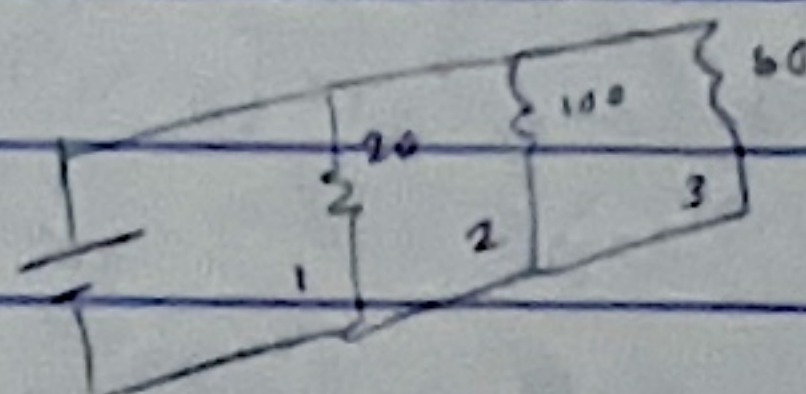
$$= 62.5V$$

5) Power dissipated through each resistor
 $P = VI$

$$\text{For } 20\Omega, 25 \times 1.25 \\ = 31.25W$$

$$\text{For } 30\Omega, 37.5 \times 1.25 \\ = 46.875W$$

$$\text{For } 50\Omega, 62.5 \times 1.25 \\ = 78.125W$$



b)

1) Total resistance in ohms

Circuit in parallel, $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$= \frac{1}{20} + \frac{1}{100} + \frac{1}{50} \quad / \quad 0.05 + 0.01 + 0.02$$

$$\frac{1}{R_T} = 0.08$$

$$\therefore R_T = \frac{1}{0.08} = 12.5\Omega$$

2) Total Current from power supply

$$I = \frac{V}{R}, \text{ where } V = 125 \\ R = 12.5$$

$$= \frac{125}{12.5} = 10A$$

3) Current through each resistor

In parallel, $I = \frac{V}{R}$

$$\text{For } 20\Omega, \frac{125}{20} = 6.25A$$

$$\text{For } 100\Omega, \frac{125}{100} = 1.25A$$

$$\text{For } 50\Omega, \frac{125}{50} = 2.5A$$

4) Voltage drop through each resistor
 $V_1 = V_2 = V_3 = V_T$ in parallel

$\therefore$ For all resistors, $V = 125\text{ V}$

5) Power dissipated through each resistor
 $P = VI$

For 20Ω , 125×6.25
 $= 781.25\text{ W}$

For $100\Omega = 125 \times 1.25$
 $= 156.25\text{ W}$

For 50Ω , 125×2.5
 $= 312.5\text{ W}$